

#Pseudocode Problem 1

BEGIN

INPUT N, K

CREATE list astronauts = [1, 2, 3, ..., N]

SET currentIndex = 0

CREATE list eliminationOrder = empty

WHILE size(astronauts) > 1 DO

 eliminateIndex = (currentIndex + K - 1) MOD size(astronauts)

 eliminatedAstronaut = astronauts[eliminateIndex]

 APPEND eliminatedAstronaut TO eliminationOrder

 REMOVE astronauts[eliminateIndex] FROM astronauts

 IF eliminatedAstronaut MOD 2 == 0 THEN

 K = K + 2

 ELSE

 K = K - 1

 ENDIF

 IF K < 2 THEN

 K = 2

 ENDIF

 IF size(astronauts) > 0 THEN

 currentIndex = eliminateIndex MOD size(astronauts)

 ENDIF

ENDWHILE

lastAstronaut = astronauts[0]

OUTPUT "Urutan eliminasi: ", eliminationOrder

OUTPUT "Astronot terakhir yang bertahan: ", lastAstronaut

END

#Pseudocode Problem 2

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BEGIN
    INPUT originalMessage
    n = LENGTH(originalMessage)
    CREATE list newLetters = empty
    newLetters[0] = originalMessage[0]
    FOR i = 1 TO n-1 DO
        currentOriginalValue = LETTER_VALUE(originalMessage[i])
        ..., Z=26
        previousOriginalValue = LETTER_VALUE(originalMessage[i-1]) //
IMPORTANT: original letter before it, not the new/ciphered one
        newValue = currentOriginalValue + previousOriginalValue
        // If the result exceeds Z (26), wrap back around to A
        WHILE newValue > 26 DO
            newValue = newValue - 26
        ENDWHILE
        newLetters[i] = VALUE_TO_LETTER(newValue)
        ..., 26=Z
    ENDFOR
    encryptedMessage = JOIN(newLetters)
    OUTPUT "Encrypted message: ", encryptedMessage
END

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